

CS443 - Network Security Assignment 6: Network Telescope Analysis

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This report shows the results of analyzing network traffic from a telescope sensor. The goal is to find scanning behavior, top scanner IPs, and protocol use. The analysis was done using Python scripts on the provided PCAP files.

1 Top Scanners

Table 1 shows the ten most active scanning sources by packet volume. These IPs generated over 9% of total traffic, with 185.242.226.50 contributing 188,000+ packets, indicating automated scanning tools or botnet activity.

IP Address	Packets	Share
185.242.226.50	188,293	2.32%
185.242.226.46	105,263	1.30%
45.146.224.132	74,096	0.91%
10.14.176.3	72,407	0.89%
10.241.128.246	72,326	0.89%
80.75.212.37	59,683	0.74%
183.136.225.42	56,189	0.69%
212.70.149.138	54,256	0.67%
118.123.105.92	51,992	0.64%
218.92.0.99	49,420	0.61%

Table 1: Top 10 scanner IP addresses by packet count

2 Target Port Analysis

Table 2 lists the top destination ports targeted by TCP and UDP scanners. Most TCP activity focused on ports commonly associated with remote access or web services (e.g., 22, 80, 443), while UDP scanning targeted DNS, SIP, and network service discovery. These results reflect broad automated probing for known services.

Port	Packets
22	174,938
3389	101,839
80	90,764
8088	82,746
8080	80,344
8728	76,159
5555	69,330
443	53,354
8122	51,620
12103	50,686

(a) TCP Scans

Port	Packets
514	144,733
53	91,129
5060	63,392
1194	40,973
32410	27,957
123	23,558
55001	13,926
1900	13,211
6881	13,009
5353	11,856

(b) UDP Scans

Table 2: Top 10 destination ports for TCP and UDP scans

3 Protocol Breakdown

Figure 1 shows the protocol breakdown of the captured traffic. TCP dominates with 7,404,342 packets (91.4%) versus UDP’s 697,351 packets (8.6%). This reflects typical scanning patterns targeting web and remote access services.

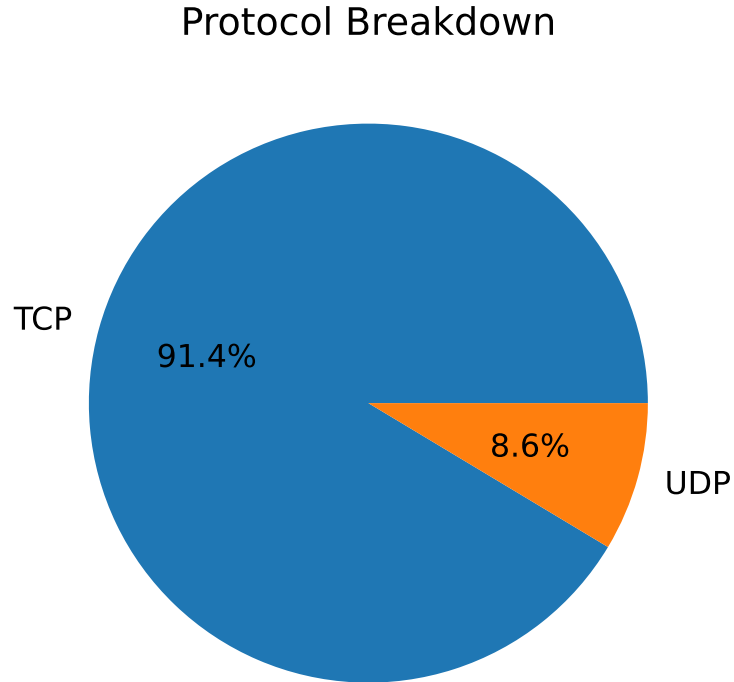


Figure 1: Protocol breakdown showing TCP vs UDP packet counts in the dataset

4 Heavy Hitter Scanner

The most prolific scanner, 88.213.209.194, originates from Bulgaria and demonstrates characteristics typical of automated reconnaissance tools. This source generated 7,598 packets targeting 832 unique IP addresses across only three destination ports, indicating a focused scanning strategy.

The limited port range suggests this scanner employs a targeted approach rather than comprehensive port sweeping, possibly searching for specific vulnerable services or devices. The high IP address diversity (832 unique targets) combined with the concentrated port selection indicates an efficient scanning methodology designed to maximize coverage while minimizing detection.

5 Traffic Visualizations

Figure 2: Temporal distribution of scanning activity using 5-minute time bins

Figure 3: Distribution analysis of the most frequently scanned ports

Figure 4: Target diversity analysis for the most active scanning sources

6 Security Analysis and Interpretation

The observed scanning patterns reveal several concerning trends in internet-wide reconnaissance activities:

Automated Mass Scanning: The concentration of traffic from a small number of sources, combined with the consistent targeting of specific ports, strongly indicates automated scanning tools rather than manual reconnaissance. This automation enables attackers to efficiently survey large address spaces for vulnerable services.

Service-Specific Targeting: The port distribution shows attackers focusing on specific services including SSH (22), RDP (3389), SIP (5060), and various web services on non-standard ports. This targeted approach suggests attackers are searching for known vulnerable implementations or default configurations.

IoT and Infrastructure Focus: Many targeted ports (8124, 34567, 37777) correspond to IoT devices, IP cameras, and embedded systems that often have poor security practices. The prevalence of these targets indicates ongoing campaigns to build IoT botnets.

DDoS Preparation: The significant UDP scanning activity, particularly targeting DNS (53), NTP (123), and SIP (5060) services, suggests reconnaissance for potential DDoS amplification attacks. These protocols can generate large response packets, making them valuable for reflection attacks.

Persistence and Scale: The steady traffic pattern over time, as shown in [Figure 2](#), indicates continuous scanning operations rather than sporadic attacks. This persistence suggests well-resourced threat actors operating scanning infrastructure.

7 Threat Assessment

While network telescopes themselves face no direct risk from this traffic (being non-routable), the observed patterns indicate active threat reconnaissance that poses risks to legitimate internet infrastructure:

- **High Risk:** The scale and persistence of scanning suggests professional threat actors
- **Botnet Activity:** Heavy hitters like 88.213.209.194 likely represent compromised hosts in larger botnets
- **Service Targeting:** Focus on specific ports indicates knowledge of vulnerable services and exploitation techniques
- **Amplification Potential:** UDP port targeting reveals preparation for large-scale DDoS attacks

Organizations should monitor for similar scanning patterns against their networks and ensure proper security controls are in place for the identified high-risk services.